auction website... It's as if Target owned Visa....[PayPal] will get cut to pieces by Amazon Payments, or by others such as Apple and by startups if it continues to be part of eBay."

Bit by bit

A mere three weeks after the launch of Apple Pay, eBay announced the separation of PayPal as an independent entity on September 30, 2014. Ironically, Apple Pay had originally approached PayPal as a partner, but eBay CEO John Donahoe reportedly forced PayPal to go with Samsung's payment services plan. But by then, a lot had happened in the world of digital payments.

For starters, bitcoin had been invented. In 2008, a paper was released online under the name Satoshi Nakamoto titled 'bitcoin: A Peer-to-Peer Electronic Cash System'. This paper detailed a peer-to-peer network for money transfer that did not rely on trust. Within one year, bitcoins were officially being issued and by 2011 services had started accepting payments via bitcoin. Bitcoin, as a mode of digital payment, never looked back from that point and on March 2, 2017, the price of one bitcoin crossed the spot price of one ounce of gold.

Bitcoin has seen a tremendous rise in value

Why did bitcoin succeed? For starters, the internet loves something that is for and by peers, instead of corporations. A typical bitcoin transaction is verified by six computers, hence it is absolutely secure. This verification process, carried out by network computers, is called bitcoin mining, where users are rewarded for helping with the transaction verification against the blockchain (a network ledger). Other factors affecting bitcoin's success are online access, reduced chance of identity theft, protection from fraud, and lower fees.

The great wall(et) of Near Fields

Two technologies transformed the way we look at digital payments in the 21st century – digital wallets and contactless payments. NFC, as a technology, was being worked on since the dawn of the 21st century. But it was the introduction of Apple Pay in 2014 that pushed NFC to the mainstream. Just scanning a credit card added it as a payment option to Apple Pay. It also added a layer of biometric authentication to it with the fingerprint scanner. This second authentication factor has in fact helped it cross transaction limits applied to one-factor transactions in several countries, like the U.K. The service is compatible with the iPhone 6, 6 Plus, iPhone 6S, 6S Plus, 7, 7 Plus, iPhone SE, iPad Air 2, iPad Pro and the Apple Watch. Users with an iPhone 5, 5C, 5S, 6, 6 Plus, 6S, 6S Plus, 7, 7 Plus and iPhone SE can use the service through an Apple Watch, though it lacks Touch ID security.

While much of the global digital payment wallets are split among key players, India has a clear leader – Paytm. While Apple Pay was launched in September 2014, Paytm wallet in India was launched almost one year earlier in 2013. And since then a lot of other mobile wallets have also emerged in India, namely Freecharge, Mobikwik, Citrus and more. While their services are differentiated mainly by the charges they levy on transactions and bank transfers, Paytm is currently the market leader in this segment. A big factor in the current popularity of Paytm was the demonetisation of ₹500 and ₹1000 notes in India on Nov 9, 2016. App downloads soared and so did transaction volumes. The Prime Minister himself was featured on a promotional ad for Paytm (which is currently under dispute).

Apple Pay is being rolled out globally and currently is available in 16 countries with several leading retailers as part of their platform.

The sound of money

While India is still behind the global market in terms of newer payment technologies like NFC, it is definitely one of the biggest digital payment economies in the world in terms of number of users. Simple features such as payment to a mobile number or scanning in-store QR codes for payment are gradually picking up pace and have become quite popular since November 2016. Innovative startups like Tonetag, which provides payment solutions with the help of sound, are constantly redefining what digital payments can do. The future of digital payments lies in novel and seamless payment methods like paying via wearables, biometric payments and mass implementation of technology like blockchain.

Currently, digital payments account for a mere 6% of total retail transactions worldwide. But with digital payments increasing everyday in volume and scope, trends indicate a much different scenario roughly 8-10 years down the line. With key merchants participating, proper IoT implementations and strong security in place, overall share could go up to 26% with 13% of that being proximity based payments, according to a BCG report. The day that happens, there will still be a few things that money can't buy. But for everything else, there will be your biometrics.